

PAPER_745: Crab Nebula MUGE — Expanding Supernova Remnant with Pulsar Wind

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Framework: Universal Quantum Field Superconductive Framework (UQFF)

Session: 180 continuation | v5.38

Date: 2025

CP4 Class: #329 — CrabNebulaExpandingMUGECalculator

Abstract

The Crab Nebula (M1) is the remnant of SN 1054, powered by a central pulsar (PSR B0531+21) at the highest rotational energy loss in the known neutron star population. Its expanding ejecta ($r(t) = r_0 + v_r \cdot t$) creates a time-dependent gravitational environment unlike any static system. This paper derives the Crab Nebula MUGE incorporating: the expanding radius $r(t)$, the pulsar wind term F_{wind} , and the nebular magnetic energy M_{mag} . The UQFF framework captures both the mechanical expansion and the magnetospheric energy injection that power the synchrotron nebula.

1. Introduction

The Crab Nebula is a pulsar wind nebula (PWN) — one of the brightest objects in X-rays and gamma-rays. The central pulsar ($P = 33$ ms, $\dot{P} = 4.2 \times 10^{-13}$ s/s) injects $\sim 5 \times 10^{31}$ W of spin-down power into the nebulae through a relativistic wind. The expanding ejecta shell moves at $v_r \sim 1500$ km/s, with angular size growing observably over human timescales. This expanding geometry requires $r(t)$ to appear explicitly in the MUGE denominator, making the Crab the canonical example of a time-dependent gravitational radius.

Crab Nebula parameters: - Distance = 2.0 kpc - Age ≈ 970 yr (SN 1054) - r_0 (initial remnant radius) $\approx 3 \times 10^{15}$ m - $r_{current} \approx 5.6$ ly = 5.3×10^{16} m - $v_r = 1500$ km/s = 1.5×10^6 m/s - $M_{ejecta} \approx 4\text{--}5 M_\odot$ - $B_{nebula} \approx 10^{-4}$ T (filamentary magnetic field) - $L_{pulsar} = 5 \times 10^{31}$ W (spin-down luminosity) - $P_{spin} = 33$ ms

2. Crab Nebula MUGE

$$\begin{aligned} g_{Crab}(r,t) = & (G \cdot M) / r(t)^2 \cdot (1 + H(z) \cdot t) \cdot (1 - B(t) / B_{crit}) \\ & + (U_{g1} + U_{g2} + U_{g3} + U_{g4}) \\ & + U_i \\ & + (\Lambda \cdot c^2 / 3) \\ & + (\hbar / \sqrt{\Delta x \cdot \Delta p}) \cdot \int (\psi \cdot H \cdot \psi \, dV) \cdot (2\pi / t_{Hubble}) \\ & + \rho_{ejecta} \cdot V \cdot g \\ & + F_{wind} \quad \quad \quad \text{[pulsar wind -- NEW]} \\ & + M_{mag} \quad \quad \quad \text{[magnetic energy -- NEW]} \end{aligned}$$

3. Time-Dependent Radius $r(t)$

The expanding radius is the defining feature of the Crab MUGE:

$$r(t) = r_0 + v_r \cdot t$$

r_0 = initial radius at SN explosion (m)
 v_r = expansion velocity = 1.5×10^6 m/s
 t = time since SN 1054 (s)

For $t = 970 \text{ yr} = 3.06 \times 10^{10} \text{ s}$:

$$r(970 \text{ yr}) = 3 \times 10^{15} + 1.5 \times 10^6 \times 3.06 \times 10^{10}$$

$$r(970 \text{ yr}) \approx 4.6 \times 10^{16} \text{ m} \approx 4.9 \text{ ly}$$

Gravitational deceleration:

$$g_{\text{grav}}(t) = G \cdot M_{\text{ejecta}} / r(t)^2$$

$$g_{\text{grav}}(970 \text{ yr}) = 6.674 \times 10^{-11} \times (4 \times 4 \times 1.989 \times 10^{30}) / (4.6 \times 10^{16})^2$$

$$g_{\text{grav}} \approx 5 \times 10^{-22} \text{ m/s}^2 \quad (\text{extremely weak} - \text{expansion dominates})$$

4. F_{wind} — Pulsar Wind Term

The pulsar wind carries relativistic particles outward, creating an inward reaction force equivalent to:

$$F_{\text{wind}} = L_{\text{pulsar}} / (4 \cdot \pi \cdot r(t)^2 \cdot c \cdot M_{\text{ejecta}})$$

$L_{\text{pulsar}} = 5 \times 10^{31} \text{ W}$
 $r(t)$ = current nebula radius
 $c = 3 \times 10^8 \text{ m/s}$

$$F_{\text{wind}} = 5 \times 10^{31} / (4 \cdot \pi \cdot (4.6 \times 10^{16})^2 \cdot 3 \times 10^8 \cdot 4 \times 1.989 \times 10^{30})$$

$$F_{\text{wind}} \approx 4.8 \times 10^{-30} \text{ m/s}^2 \quad (\text{small but physically significant over 970 yr})$$

Cumulative momentum injection over nebula lifetime:

$$\Delta p_{\text{wind}} = L_{\text{pulsar}} \cdot t / c \approx 5 \times 10^{31} \times 3 \times 10^{10} / 3 \times 10^8 \approx 5 \times 10^{33} \text{ kg} \cdot \text{m/s}$$

5. M_{mag} — Magnetic Energy Term

The nebular magnetic field stores and dissipates energy, affecting dynamics:

$$M_{\text{mag}} = B^2 / (2 \cdot \mu_0 \cdot r(t) \cdot \rho_{\text{ejecta}}) \quad [\text{magnetic pressure divided by density}]$$

$B = 10^{-4} \text{ T}$ (filamentary field)
 $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
 $\rho_{\text{ejecta}} \approx 10^{-22} \text{ kg/m}^3$ (current)

$$M_{\text{mag}} = (10^{-4})^2 / (2 \cdot 4\pi \times 10^{-7} \cdot 4.6 \times 10^{16} \cdot 10^{-22})$$

$$M_{\text{mag}} \approx 2.7 \times 10^{-17} \text{ m/s}^2$$

This is the dominant non-Newtonian term for the Crab, controlling the synchrotron brightness evolution.

6. UQFF Gravity Components

U_{g1} : Pulsar magnetic dipole
 $\mu_{\text{dipole_PSR}} \approx 3.8 \times 10^{30} \text{ J/T}$ (from P , $\dot{P}$)

B_PSR = 3.8×10^8 T (surface field)
U_g1 contributes at pulsar vicinity only

U_g2: Superconductive aether field threading nebula
B_super aligned with pulsar rotation axis
 $U_{g2} \approx 10^5$ J/m³ (strong near pulsar wind shock)

U_g3: External galactic field at 2 kpc
 $U_{g3} = G \cdot M_{gal} / r_{gal}^2$

U_g4: Galactic center contribution
 $k_4 \cdot \rho_{vac, [SCm]} \cdot (M_{bh}/d_g) \cdot e^{(-\alpha t)} \cdot \cos(\pi \cdot t_n)$

7. Temporal Evolution Summary

Time (yr)	r(t) (ly)	g_grav (m/s ²)	M_mag (m/s ²)
0	0.3	2×10^{-14}	10^{-16}
100	1.6	5×10^{-21}	10^{-17}
970	4.9	5×10^{-22}	2.7×10^{-17}
10,000	~50	$\sim 5 \times 10^{-27}$	$\sim 10^{-19}$

The magnetic term M_mag remains important relative to g_grav throughout nebula evolution.

8. Conclusion

The Crab Nebula MUGE demonstrates that time-dependent radius r(t) is essential for accurately modeling supernova remnant gravity. The F_wind pulsar injection and M_mag magnetic energy terms together dominate over classical Newtonian gravity within the nebula. The UQFF successfully models the transition from SN ejecta to mature PWN through its modular environmental forcing framework.

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